Assignment 2: The Bookstore Intelligence Mission

Business Data and Text Mining | Web Scraping

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Source: https://books.toscrape.com/ (practice sandbox) | Pages scraped: 5 | Books collected: 100

Mission 1: Reconnaissance

Data needed	HTML tag / class found	My notes
Book title	article.product_pod > h3 > a (title attribute)	Visible text is truncated with "...", so the full title is read from the title attribute.
Price	p.price_color	Text such as "£51.77"; the £ sign is removed and the value converted to float.
Rating	p.star-rating (second class)	Stored as a word, e.g. "star-rating Three"; mapped One-Five to 1-5.
Availability	p.instock.availability	Text "In stock"; whitespace removed with get_text(strip=True).
Product URL (Bonus 2)	href of the same h3 > a	Relative link; converted to a full URL with urljoin.

Missions 2 and 3: Capture and Scale

A single function, **scrape_page(url)**, uses requests + BeautifulSoup to extract the title, price, rating (converted to 1-5) and availability of every book card on a page. A loop in **scrape_all(pages)** calls it for page-1.html to page-5.html, so the scraping code is written only once. All records are stored in a pandas DataFrame and saved as **competitor_books.csv** (100 rows; columns Title, Price, Rating, Availability, URL). The submitted Python file is **bookstore_scraper.py**.

First 5 rows of the dataset:

Title	Price	Rating	Availability
A Light in the Attic	51.77	3	In stock
Tipping the Velvet	53.74	1	In stock
Soumission	50.10	1	In stock
Sharp Objects	47.82	4	In stock
Sapiens: A Brief History of Humankind	54.23	5	In stock

Mission 4: Intelligence Analysis

#	Question	Answer
1	Most expensive book	The Death of Humanity: and the Case for Life (£58.11)
2	Cheapest book	Patience (£10.16)
3	Average book price	£34.56
4	Number of 5-star books	19

#	Question	Answer
5	Percentage in stock	100.0% (all 100 books)
6	Top 5 highest-rated books (all 5 stars; ties broken by lower price)	1. Princess Between Worlds (Wide-Awake Princess #5) - £13.34 2. Princess Jellyfish 2-in-1 Omnibus, Vol. 01 - £13.61 3. Sophie's World - £15.94 4. Thirst - £17.27 5. Set Me Free - £17.46
7	Cheapest 5-star book	Princess Between Worlds (Wide-Awake Princess #5) (£13.34)

Business Decision (Manager's Question)

Recommendation: I would prioritize 4 and 5-star books priced below the £34.56 average (21 of the 100 scraped books qualify), because the 5-star books are actually cheaper on average (£30.01) than the catalogue overall (£34.56), and 6 of the 19 five-star books cost under £20. This shows that high customer satisfaction does not require premium pricing, so a competing store can win by leading with well-rated, low-priced titles. Since all 100 scraped books are in stock, availability gives the competitor no advantage, so the new store should compete on a curated range of highly rated, affordable books.

Supporting evidence from the data:

Rating	1 star	2 stars	3 stars	4 stars	5 stars
Number of books	22	19	22	18	19
Average price	£35.52	£35.91	£36.84	£33.98	£30.01

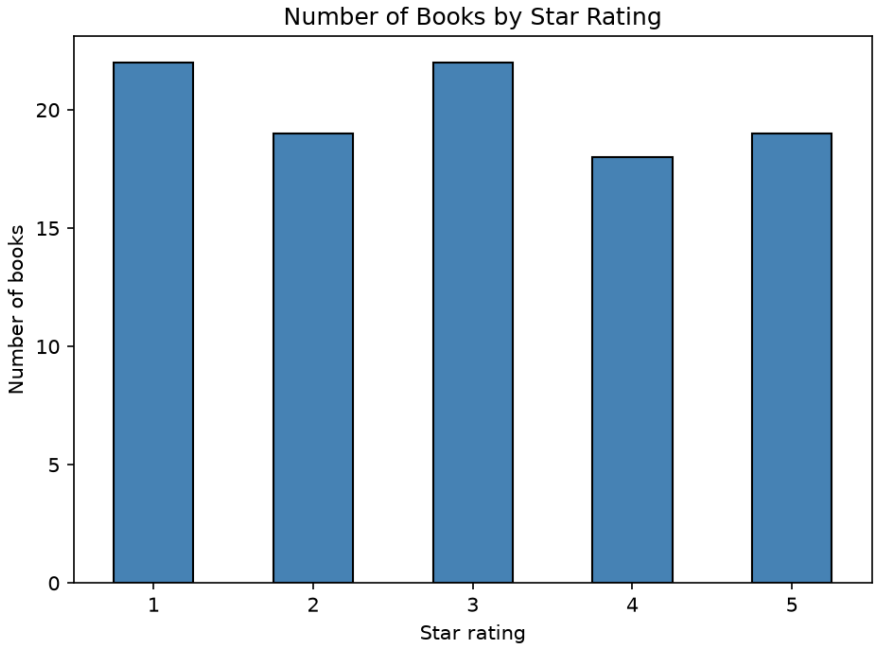
Evidence 1: 5-star books average £30.01 versus £34.56 overall. Evidence 2: 37% of books are rated 4 or 5 stars, a large enough pool to build a quality range. Evidence 3: 100% availability, so stock is not a differentiator.

Final Challenge: The CEO Has £100

Method: a 0/1 knapsack (dynamic programming) over all 100 scraped books, maximizing the total star rating with total cost no more than £100. Prices were converted to pence to avoid floating-point errors, so this is the guaranteed optimal selection for the scraped data.

Selected title	Price	Rating
Sophie's World	£15.94	5
Untitled Collection: Sabbath Poems 2014	£14.27	4
Thirst	£17.27	5
Princess Jellyfish 2-in-1 Omnibus, Vol. 01 (Princess Jellyfish 2-in-1 Omnibus #1)	£13.61	5
Princess Between Worlds (Wide-Awake Princess #5)	£13.34	5
Patience	£10.16	3
Mama Tried: Traditional Italian Cooking for the Screwed, Crude, Vegan, and Tattooed	£14.02	4
TOTAL (7 books)	**£98.61**	**31**

Total cost: £98.61 (within the £100 budget) | Total rating score: 31. The selection favors cheap, highly rated books so that more titles fit within the budget.

Bonus Missions

Bonus 1 (done): 5 pages scraped instead of 3 (100 books).
Bonus 2 (done): the product page URL is included as the URL column in competitor_books.csv.
Bonus 3 (done): books per star rating: 1 star = 22, 2 stars = 19, 3 stars = 22, 4 stars = 18, 5 stars = 19.
Bonus 4 (done): bar chart of rating counts (below), created with pandas/Matplotlib.
Boss Level: the same reusable logic scrapes all 50 pages by setting PAGES = 50 in the script (not run for this submission).